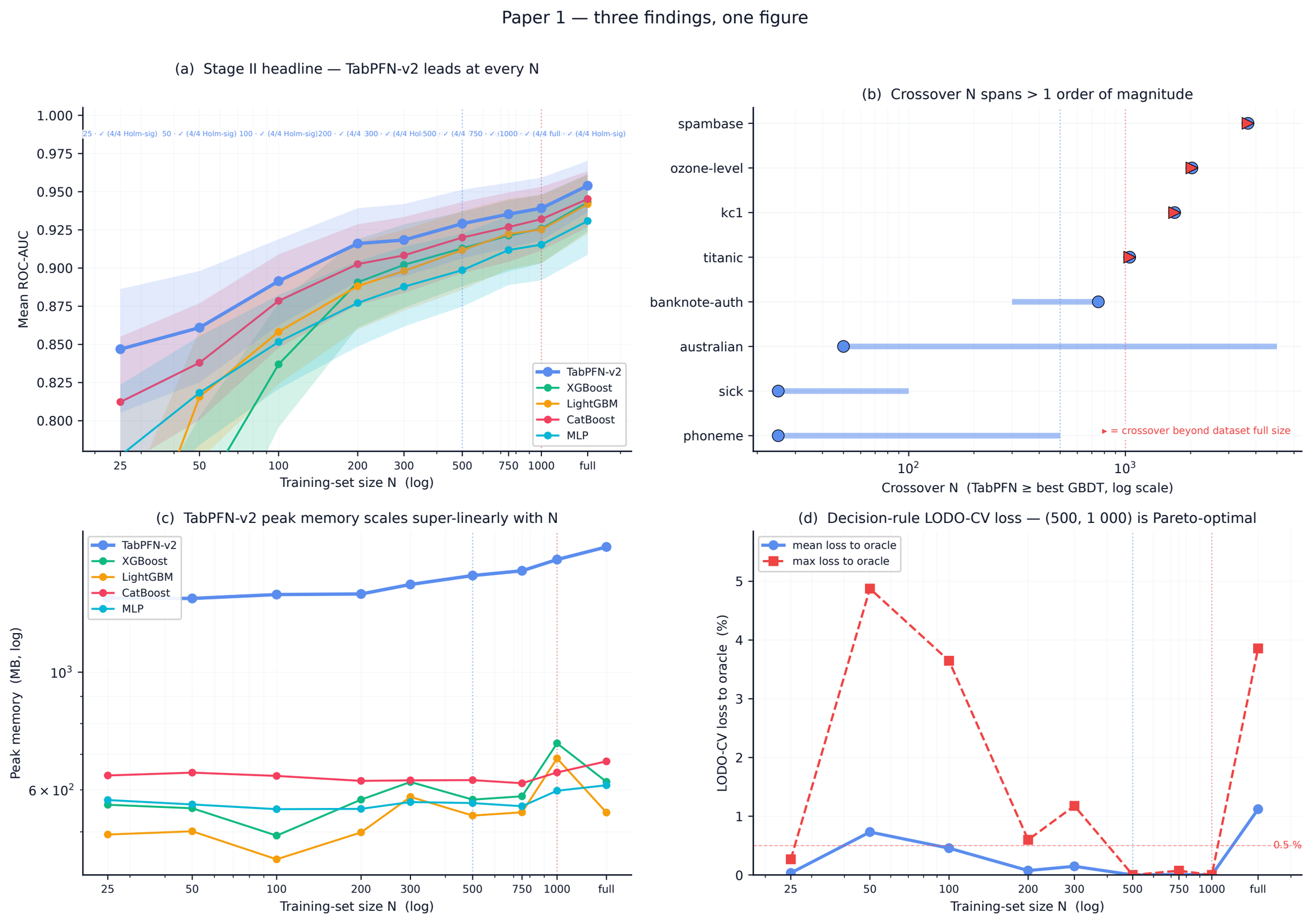

Paper 1 — three findings, one figure
(a) Stage II headline — TabPFN-v2 leads at every N
Mean ROC-AUC
1.000
0.975
0.950
0.925
0.900
0.875
0.850
0.825
0.800
25 · ✓ (4/4 Holm-sig) 50 · ✓ (4/4 Holm-sig) 100 · ✓ (4/4 Holm-sig) 200 · ✓ (4/4 300 · ✓ (4/4 Hol 500 · ✓ (4/4 750 · ✓ (1000 · ✓ (4/4 full · ✓ (4/4 Holm-sig)
TabPFN-v2
XGBoost
LightGBM
CatBoost
MLP
25
50
100
200
300
500
750
1000
full
Training-set size N (log)
(b) Crossover N spans > 1 order of magnitude
spambase
ozone-level
kc1
titanic
banknote-auth
australian
sick
phoneme
▸ = crossover beyond dataset full size
10^2
10^3
Crossover N (TabPFN ≥ best GBDT, log scale)
(c) TabPFN-v2 peak memory scales super-linearly with N
Peak memory (MB, log)
10^3
6×10^2
TabPFN-v2
XGBoost
LightGBM
CatBoost
MLP
25
50
100
200
300
500
750
1000
full
Training-set size N (log)
(d) Decision-rule LODO-CV loss — (500, 1 000) is Pareto-optimal
LODO-CV loss to oracle (%)
0
1
2
3
4
5
mean loss to oracle
max loss to oracle
0.5 %
25
50
100
200
300
500
750
1000
full
Training-set size N (log)